



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 12



DEUTZ-FAHR FOUR-WHEEL TRACTOR
Model : AGROLUX 95

DEUTZ-FAHR AGROLUX 95 FOUR-WHEEL TRACTOR

Test Report No	:	2013 - 12
Date of Test	:	February 15-16, 2013
Test Valid Until	:	February 16, 2018
Brand	:	DEUTZ-FAHR
Model	:	AGROLUX 95
Serial No.	:	AG95E1WUT17288
Type	:	Four-Wheel Drive Tractor
Manufacturer	:	SAME DEUTZ-FAHR ITALIA S.p.A. Viale F. Cassani, 15, 24047 treviglio (BG), Italy
Test Requested by	:	HEXAGON DISTRIBUTING CORPORATION G/F SGS Foundation Building, 1335 G. Araneta Avenue, Quezon City
Machine Classification	:	Commercial

SUMMARY

The DEUTZ-FAHR AGROLUX 95 Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

The Maximum Power developed at the Power-Take-Off (PTO) shaft of the tractor was 61.5 kW at 561 rpm PTO Shaft Speed. The Engine Crankshaft Speed at this point was 2161 rpm with Fuel Consumption Rate of 20.4 L/h. The PTO Shaft Torque at this point was 106.8 kg-m. The Maximum PTO Shaft Torque was 118.9 kg-m which occurred at 419 rpm PTO Shaft Speed.

The Radii of Turning Circle of the tractor when making a right and left turns were both 4970 mm without brake. The Radii of turning circle of the tractor with brake was not conducted, because of the presence of front wheel brake. When brake was applied on the left rear wheel the front wheel also stopped. Likewise, when brake was applied in the right rear wheel, the front wheel also stopped. During test the tractor cannot make a turn.

The Radii of Turning Area of the tractor when making a right and left turns were both 5160 mm without brake.

The Travelling Speeds of the tractor at the engine full throttle speed setting (Pre-set) recommended by the manufacturer/Applicant of test at different gear settings were the following: For the FORWARD travelling speeds, at “LOW” speed range the forward travelling Speeds were 1.26 kph, 1.67 kph, 2.21 kph, 3.20 kph, and 4.31 kph at 1st, 2nd, 3rd, 4th, and 5th gear settings, respectively, at “MEDIUM” speed range the forward travelling Speeds were 3.98 kph, 5.38 kph, 6.98 kph, 10.0 kph, and 13.7 kph at 1st, 2nd, 3rd, 4th, and 5th gear settings, respectively. At “HIGH” speed range the forward travelling Speeds were 11.7 kph, 15.6 kph, 20.7 kph, 29.6 kph, and 37.0 kph at 1st, 2nd, 3rd, 4th, and 5th gear settings, respectively. For the REVERSE travelling speeds, at “LOW” speed range the forward travelling Speeds were 1.30 kph, 1.77 kph, 2.27 kph, 3.30 kph, and 4.44 kph at 1st, 2nd, 3rd, 4th, and 5th gear settings, respectively, at “MEDIUM” speed range the forward travelling Speeds were 4.16 kph, 5.58 kph, 7.44 kph, 10.6 kph, and 14.4 kph at 1st, 2nd, 3rd, 4th, and 5th gear settings, respectively. At “HIGH” speed range the forward travelling Speeds were 12.3 kph, 16.8 kph, 23.0 kph, 31.4 kph, and 38.3 kph at 1st, 2nd, 3rd, 4th, and 5th gear settings, respectively.

SCOPE OF TEST

The tractor was tested to establish its performance characteristics. The test was conducted from February 15 to 16, 2013 at the AMTEC Test Laboratory, UPLB, College, Laguna. It was evaluated on the following items:

1. Suitability of operator's manual;
2. Verification of Specifications
3. Laboratory tests
 - a. PTO load performance test;
 - b. Determination of Turning Circle
 - c. Determination of Turning Area
 - d. Determination of Travelling Speeds

METHOD OF TEST

The tractor was tested based on Philippine Agricultural Engineering Standards, PAES 119:2001 Agricultural Machinery – Four-Wheel Tractor – Methods of Test.

DESCRIPTION OF THE TRACTOR

The DEUTZ-FAHR AGROLUX 95 Four-wheel tractor(**Figure 1**) was a 67.5 kW four-wheel drive tractor (as stated in the operator's manual).

The tractor used a four-stroke, water-cooled, turbocharged, four cylinder diesel engine. Fuel feed system was a force-feed type using a circulation pump. The fuel tank (**Figure 2**) was located just at the back of the operator's seat. It utilized a dry dual type air cleaner.(**Figure 3**). Lubrication system was force-feed lubricating type. Cooling was done through circulation of water from cylinder jacket to the

radiator using a circulation pump. It was started by an electric starting system. A vertical muffler was used in the exhaust system.

The tractor's transmission system had 15 forward speeds and 15 reverse speeds, which were grouped into synchro-shuttle setting, three-range setting, and five main gear settings. The synchro-shuttle setting were forward and reverse, the range settings were "LOW", "MEDIUM", and "HIGH" while the main gear settings were 1st, 2nd, 3rd, 4th, and 5th. The Main gear was fully synchronized while the range gear could only be shifted when the tractor is completely stopped. The synchro-shuttle, range, and main gear settings could be set through levers at the right side of operator's seat. A parking brake was also provided and can be actuated through a lever at the left side of the operator's seat.

The tractor's direction was being controlled through a hydrostatic power steering system with the steering wheel (**Figure 4**) located in front of the driver's seat. The brakes of the rear wheels were independent of each wheel, however, they could be locked-on through the pedals. Also, the front wheels were installed with brake system simultaneous with the rear wheels. That is, the left rear and left front wheels were braked together the same way with the right rear and right front wheel. The tractor used a hydraulic brake system. A differential lock could be set by foot pedals located on the tractor's flooring just underneath the operator's seat. The size of the rear wheel tires were 46.0 cm x 76.2 cm (460/85 R30) and the front wheel tires were 38.0 cm x 61.0 cm (380/85 R24). Both have spikes.

The tractor had a three-point hitch linkage (**Figure 5**) for mounted implements. It had a swinging drawbars (**Figure 6a**) for pulling a trailer or any trailing implement.

The Power-Take-Off shaft (**Figure 6b**) of the tractor was located at the rear of the tractor and was set at 540 rpm. The PTO shaft could be engaged/disengaged by pulling/pushing a handle located at the right side of the operator seat.

The tractor had two headlights, two brake lights, four turning lights and one working light.

The operator's seat (**Figure 7**) was adjustable for operator's comfort.

The tractor controls (**Figure 8**) were located in front of and beside the operator's seat and included the following: steering wheel, clutch pedal, hand throttle lever, ignition key, main gear shift lever, range gear shift lever, synchro-shuttle lever, differential lock pedals, brake pedals, foot throttle pedal, PTO controls, hydraulic lift position control lever and auxiliary hydraulic control lever. The panel board (**Figure 8**) which was also located in front of the operator's seat contained the following: tachometer, fuel gage, hour meter, coolant temperature gage, PTO speed indicator, speedometer, air filter clogged indicator light, engine pre-heating indicator light, battery charge indicator light, hydraulic oil pressure warning light, engine oil pressure warning light, brake fluid level warning light, hydraulic oil filter warning light, general alarm warning light, PTO operation indicator light, and turning and hazard indicators light. **Table 1** shows the detailed specifications of the tractor.



Figure 1. DEUTZ-FAHR AGROLUX 95 Four-Wheel Tractor



Figure 2. Fuel tank

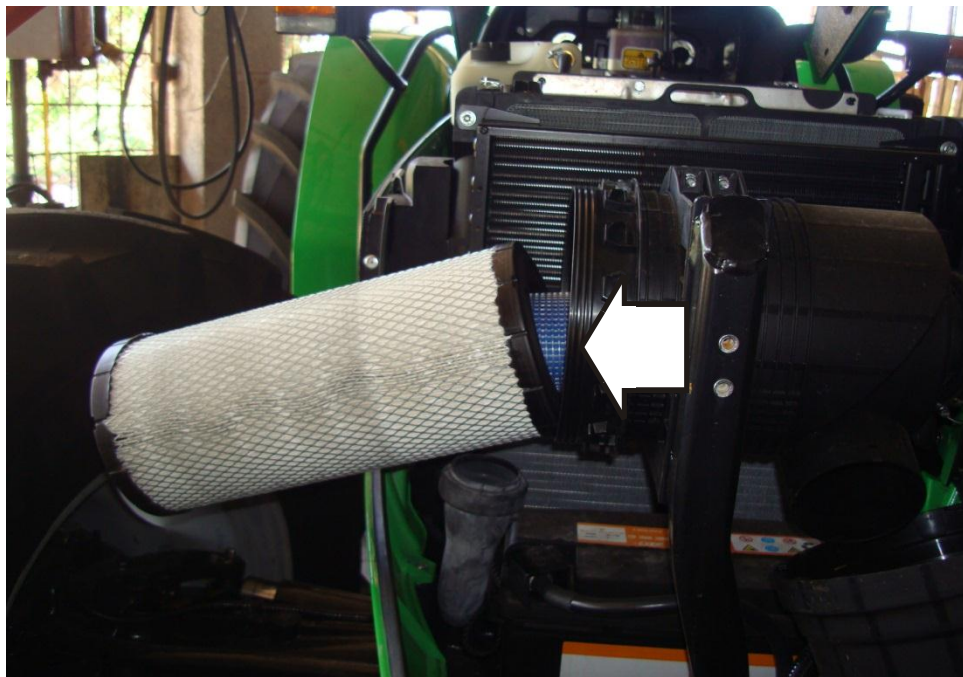


Figure 3. Air Cleaner



Figure 4. Steering wheel

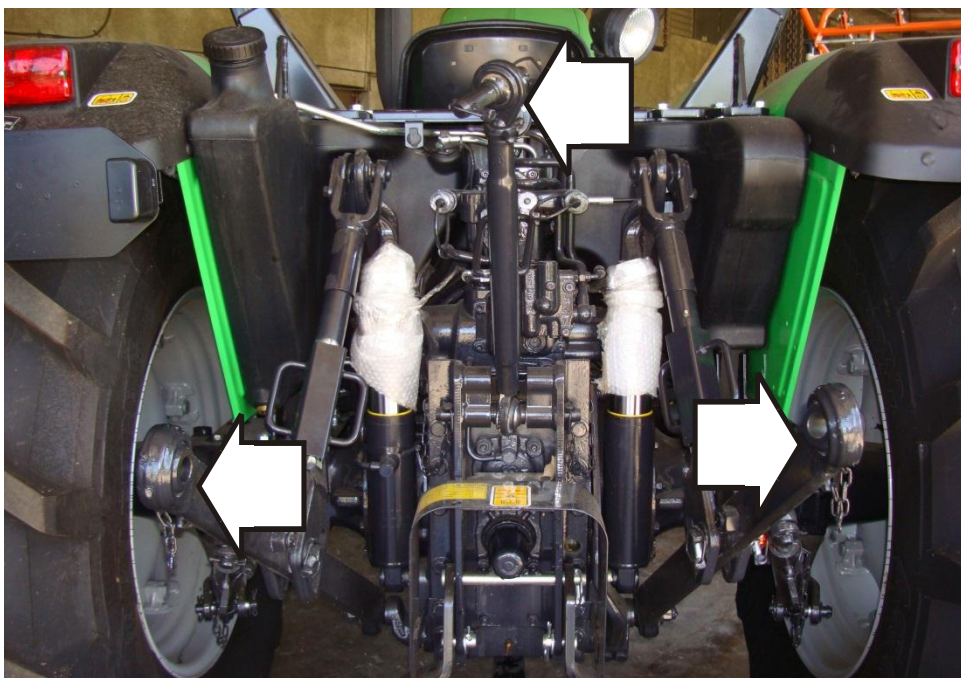


Figure 5. Three-Point Hitch

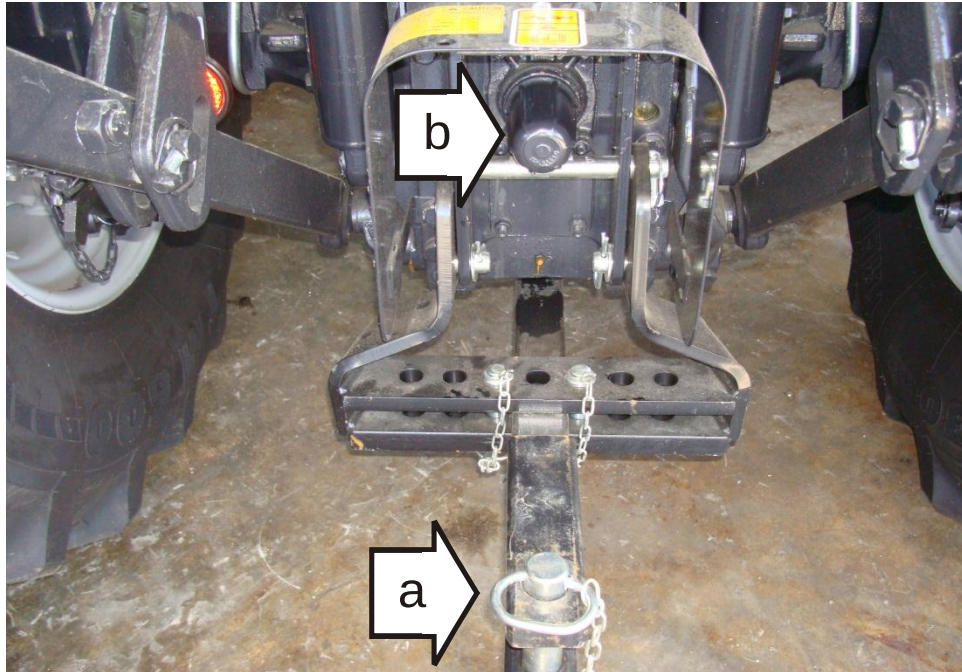


Figure 6. (a) Swinging drawbar
(b) PTO output shaft



Figure 7. Operator's seat

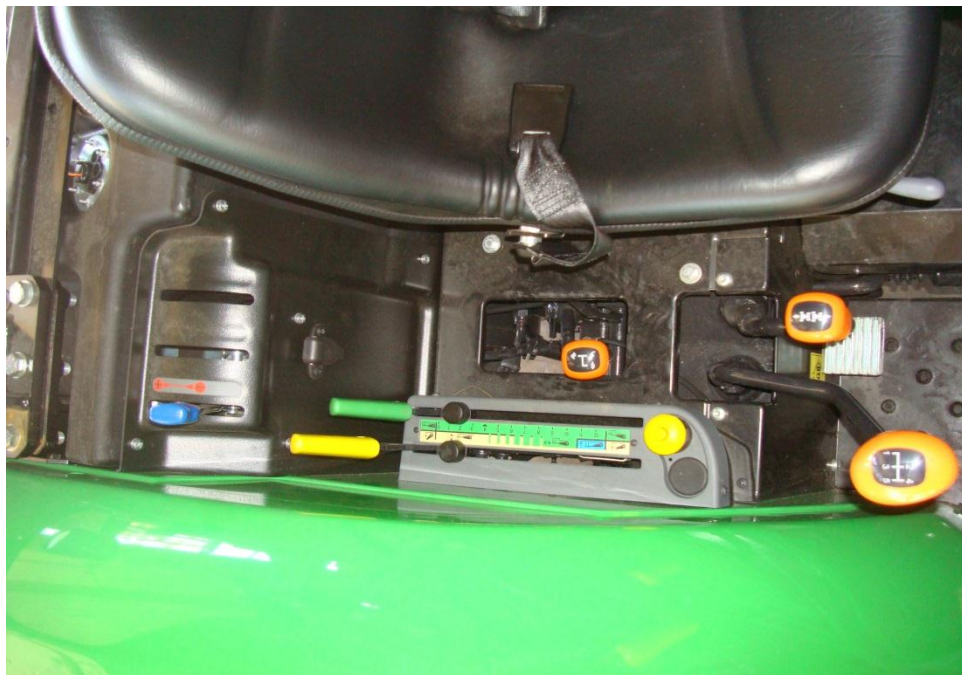


Figure 8. Tractor's control and panel board

SPECIFICATIONS

Table 1. Detailed specifications of the DEUTZ-FAHR AGROLUX 95 Four-Wheel Tractor

1.	Tractor specifications (Based on actual ocular inspection of the tractor and operator's manual submitted)		
1.1	Brand	:	DEUTZ-FAHR
1.2	Model	:	AGROLUX 95
1.3	Serial No.	:	AG95E1WUT17288
1.4	Make	:	SAME DEUTZ-FAHR ITALIA S.p.A., Treviglio, Italy
1.5	Type	:	Four-Wheel Drive
1.6	Rated Maximum Power (kW)	:	67.5
2.	Dimensions and Weight		
2.1	Overall length (mm) With front ballast	:	4460
2.2	Overall width (mm)	:	2105
2.3	Overall height (mm)	:	2555 (Canopy)
3.	Tractor's engine		
3.1	Brand	:	DEUTZ-FAHR
3.2	Type	:	Four-Stroke Cycle, Water-cooled, Turbocharged, Four Cylinder Diesel Engine
3.3	Model	:	SDF 1000.4 WT
3.4	Make	:	No data available
3.5	Serial Number	:	16491
3.6	Rated Output Power (kW)	:	67.5
3.7	Number of Cylinder	:	4

Table 1. Con't

3.7	Bore x stroke (mm)	:	No data
3.8	Total displacement volume (cc)	:	4000
3.9	Fuel system		
3.9.1	Type of fuel feed	:	Circulation pump
3.9.2	Fuel used	:	Diesel
3.10	Cooling system	:	Water-cooled
3.11	Lubricating system	:	Force-feed type
3.12	Starting system	:	Electric type using battery
3.13	Air cleaner	:	Dry dual type
3.14	Exhaust system	:	Vertical muffler
4.	Power-Take-off		
4.1	Location	:	At the rear of the tractor
4.2	Diameter of PTO shaft end (mm)	:	35.0
4.3	Number of splines	:	6
4.4	Height above ground (mm)	:	735
4.5	Rated Shaft Speed(rpm)	:	540
4.6	Direction of rotation (viewed from the rear of the tractor)	:	Clockwise
4.7	Mode of Operation	:	Lever
5.	Drawbar		
5.1	Type	:	Swinging
5.2	Diameter of hitchpoint (mm)	:	33.0
6.	Tire		
6.1	Front		
6.1.1	Number	:	2
6.1.2	Size (cm)	:	38.0 x 61.0
6.1.3	Track width (mm)	:	1535

Table 1. Con't

6.2	Rear		
6.2.1	Number	:	2
6.2.2	Size	:	46.0 x 76.2
6.2.3	Track width (mm)	:	1590
7.	Wheel base (mm)	:	2365
8.	Ground Clearance (mm)	:	362 (Drawbar bracket)

Table 2. Dimension of linkage geometry of DEUTZ-FAHR AGROLUX 95 tractor
(see also **Figure 9**)

Particulars		Data (mm)
1. Length of Lift Arms	A	315
2. Length of Lower Links	B	850
3. Horizontal distance between two Lower Link Points	u	
4. Horizontal distance between the two Lift Arm End Points	v	485
5. Distance of Lower Link Pivot Points to Lift Rod Pivot Points on Lower Links	D	465
6. Length of Lift Rods	L	645 - 988
7. Height of Lower Hitch Points relative to the rear wheel axis		
a. Low Position	h	195
b. High Position	H	245
8. Height above ground of Lower hitch points when locked in transport position		1085
9. Three-point-hitch		Category II

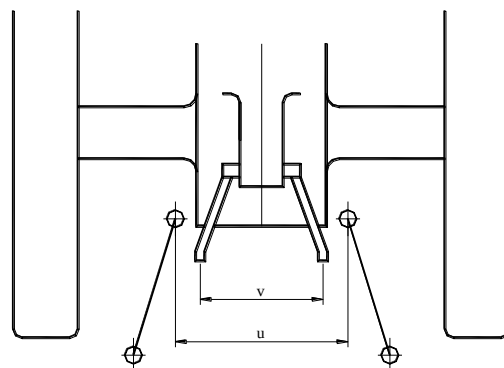
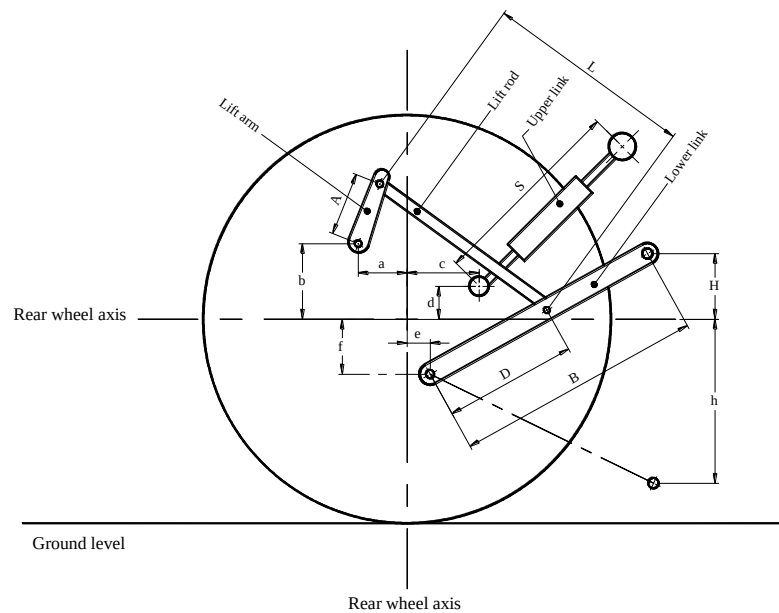


Figure 9. Linkage Geometry

OPERATOR'S MANUAL

The operator's manual submitted with the tractor was presented in English and contained the following: Safety precautions; Operator instruction (Detailed pre-operation and operation procedures); Accessories, spare parts, and consumables; Instruction on maintenance; Storage, Delivery, acceptance and transportation; Technical specifications; Disassembly and disposal; Warranty term; and Appendix.

RESULTS AND DISCUSSION

The DEUTZ-FAHR AGROLUX 95 Four-Wheel tractor was tested to establish its performance. It was tested at the original settings set by the manufacturer.

A. DETERMINATION OF TURNING CIRCLE AND TURNING AREA

The tractor was tested in a suitable concrete area. It was tested by operating the tractor at the slowest speed possible. The Steering wheel was rotated to the maximum that it can rotate during turning to the left and right directions. The Radii of Turning Circle and Turning Area with and without brakes were measured. The Radius of Turning Circle is the Radius of the smallest circle tangentially described by the median plane of the outermost wheel, while the Radius of Turning Area is the radius of the smallest circle described by the outermost point of the tractor. Results of the test are shown in **Tables 3 and 4**.

TURNING CIRCLE AND TURNING AREA

Tractor on Test : DEUTZ-FAHR
 Model : AGROLUX 95
 Date of Test : February 16, 2013

Table 3. The Radius of Turning Circle of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	4970	4970
WITH BRAKES	-	-

Table 4. The Radius of Turning Area of the tractor

	RIGHT TURN	LEFT TURN
WITHOUT BRAKES	5160	5160
WITH BRAKES	-	-

Note: The Radius of Turning Circle and Radius of Turning Area with brake was not conducted due to the presence of brakes on the front wheels. When the left rear wheel was braked, the left front wheel also was braked. The same with the right rear wheel and right front wheel. The tractor could not make a turn when braked during testing.

B. DETERMINATION OF TRAVELLING SPEEDS

The Travelling Speeds of the tractor were measured in actual condition. These were measured on a 50-meter concrete road. The tractor was operated at full throttle speed setting and made to run at different range and speed settings. The time elapsed for the tractor to cross the 50-meter distance was obtained. The fifth gear setting at range “HIGH”, yielded the fastest Forward Travelling speed of 37.0 kph, while the first gear setting at range “LOW” was slowest at 1.26 kph. For the Reverse Travelling Speed, the speed at first gear setting was 1.30 kph, while at fifth gear setting the speed was 38.3 kph. In general, the tractor speeds increased with increased range and gear settings. **Table 5** shows the results of the test.

TRAVELLING SPEEDS

Tractor on Test : DEUTZ-FAHR
 Model : AGROLUX 95
 Date of Test : February 16, 2013

Table 5. The Travelling Speeds of the Tractor in kph at different settings

SYNCHRO-SHUTTLE SETTINGS	RANGE GEAR SETTINGS	MAIN GEAR SETTINGS (kph)				
		FIRST	SECOND	THIRD	FOURTH	FIFTH
FORWARD	LOW	1.26	1.67	2.21	3.20	4.31
	MEDIUM	3.98	5.38	6.98	10.0	13.7
	HIGH	11.7	15.6	20.7	29.6	37.0
REVERSE	LOW	1.30	1.77	2.27	3.30	4.44
	MEDIUM	4.16	5.58	7.44	10.6	14.4
	HIGH	12.3	16.8	23.0	31.4	38.3

C. PTO PERFORMANCE LOAD TEST

The power at the Power-Take-Off shaft of the tractor was obtained using a Portable Waterbrake Dynamometer. It was tested at full throttle speed setting of the tractor. Data such as dynamometer load, PTO rpm, Engine rpm, fuel consumption were obtained at 20 & 40 rpm speed increments. The Maximum No-Load speed of the engine was 2472 rpm. The Maximum PTO Output Power obtained was 61.5 kW at 561 rpm PTO shaft speed. At this point, the PTO torque was 106.8 kg-m with tractor's Fuel Consumption Rate of 20.4 L/h. No breakdown and other malfunctioning occurred during the test. The test set-up is shown in **Figure 10** and the results of test are shown in **Table 6** and plotted in **Figure 11**.



Figure 10. The Tractor on the PTO test set-up

PTO PERFORMANCE TEST

Tractor on Test : DEUTZ-FAHR
 Model : AGROLUX 95
 Date of Test : February 15, 2013

Table 6. Results of PTO load performance test

SHAFT SPEED (rpm)		PTO		FUEL CONSUMPTION (L/h)	SPECIFIC FUEL CONSUMPTION (g/kW-h)
ENGINE	PTO	TORQUE (kg-m)	POWER (kW)		
2472	651	9.40	6.30	6.57	880.0
2396	622	66.3	42.3	14.9	296.9
2328	602	92.7	57.3	18.4	269.8
2233	581	102.3	61.0	20.5	282.4
2161	561	106.8	61.5	20.4	278.2
2096	541	109.0	60.6	20.0	277.8
1979	521	111.9	59.8	19.8	277.6
1814	501	114.9	59.1	19.3	274.5
1714	460	117.1	55.3	18.3	278.6
1612	419	118.9	51.1	16.7	274.8

REMARKS / OBSERVATIONS: No breakdown and any malfunctioning occurred.

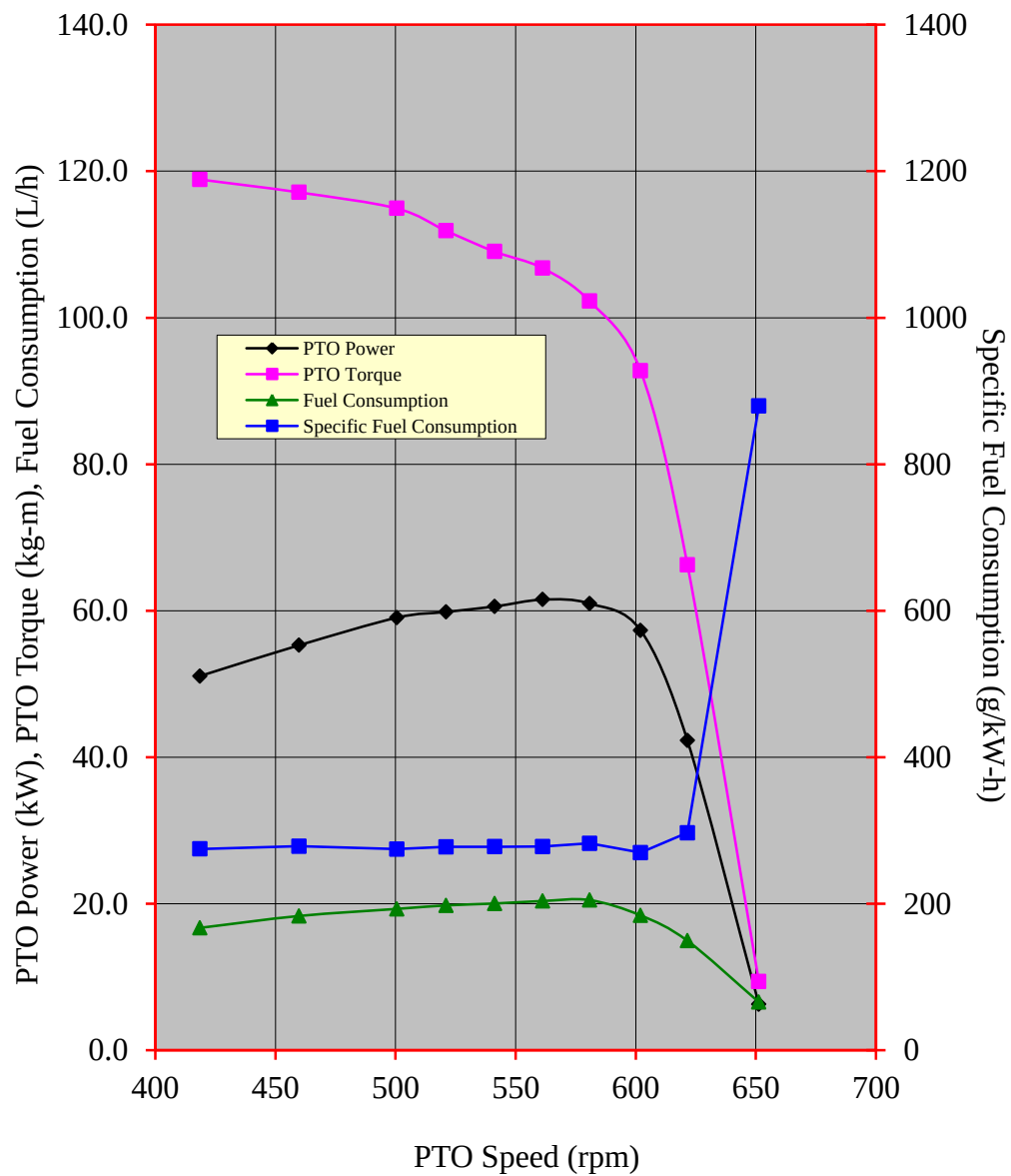


Figure 11. Performance Characteristic Curves of the
DEUTZ-FAHR AGROLUX 95 Four-Wheel Tractor

Test Engineer:

ROMULO E. EUSEBIO

Checked by:

DARWIN C. ARANGUREN

Approved for release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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